

Q-EVM: A Quantum-Resistant Account Abstraction Layer for Ethereum

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Abstract—Ethereum’s ECDSA-based transaction authorization is vulnerable to quantum attacks. While post-quantum signatures like ML-DSA offer security, their direct on-chain verification is economically infeasible. This paper presents Q-EVM, a feasibility study that combines ERC-4337 Account Abstraction with a zkVM coprocessor to enable quantum-resistant wallets without modifying Ethereum’s core protocol. Unlike prior work that advocates hard-fork precompiles, we explore the design space of forkless integration, characterizing the fundamental trade-off between off-chain proving latency and on-chain gas cost. Our key insight is that batching multiple ML-DSA verifications in a zkVM reduces amortized latency to levels competitive with existing solutions for high-value or batched transactions. We report projected performance based on existing zkVM benchmarks (e.g., RISC Zero) and identify the specific conditions under which forkless PQC becomes viable. This work is a *design space exploration and feasibility study*, not a production-ready system. The results suggest that with further optimizations (e.g., recursive proofs, dedicated provers), forkless PQC could become practical for a wider range of applications.

Index Terms—Post-Quantum Cryptography, Account Abstraction, ERC-4337, Zero-Knowledge Proofs, Blockchain Security, ML-DSA, Dilithium, Feasibility Study

I. INTRODUCTION

A. Motivation and Quantum Threat

Ethereum secures over \$400 billion in assets using digital signatures based on the secp256k1 elliptic curve. The security of these accounts hinges on the computational hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP). Shor’s quantum algorithm provides a polynomial-time solution to ECDLP, meaning a sufficiently powerful quantum computer could derive private keys from public keys and signatures, leading to catastrophic loss of funds [1].

The National Institute of Standards and Technology (NIST) has standardized Module-Lattice-Based Digital Signature Algorithm (ML-DSA, formerly Dilithium) as FIPS 204 [2]. While ML-DSA is mathematically secure against known quantum attacks, its implementation in the Ethereum Virtual Machine (EVM) is economically infeasible. Benchmarks show that verifying a single ML-DSA signature natively on-chain would consume over **2.5 million gas** (estimated from bytecode

simulation) [3], vastly exceeding the current Ethereum block gas limit of 30 million for a single verification.

B. Problem Definition and Research Gap

There exists a critical gap between cryptographic security (quantum resistance) and economic viability (gas costs). Existing proposals fall into two categories:

- 1) **Hard-fork precompiles** (e.g., EIP-8052 for Falcon): Require years of governance coordination and protocol changes.
- 2) **Native EVM verification**: Gas costs exceed block limits, making them impractical.

To the best of our knowledge, no existing work systematically explores the **forkless design space** using zkVM coprocessors for PQC, nor characterizes the trade-off between proving latency and on-chain gas cost.

C. Contributions and Novel Insights

This paper does **not** claim to present a production-ready system. Instead, we offer a feasibility study and design space exploration. Our key contributions are:

- **Insight 1 – Bottleneck identification**: The fundamental bottleneck for PQC on Ethereum is not signature size (which can be handled off-chain) but **verification cost**. We show that a zkVM can shift this cost off-chain while maintaining on-chain security via succinct proofs.
- **Insight 2 – Trade-off characterization**: We explicitly characterize the trade-off between off-chain proving latency (12.4 seconds per signature projected for a single ML-DSA verification) and on-chain gas cost (242k gas projected for Groth16 verification). This trade-off is fundamental: faster proving requires more efficient circuits or dedicated hardware, while lower gas requires succinct proofs.
- **Insight 3 – Batching as an enabler**: We discover that batching multiple ML-DSA verifications in a single zkVM execution reduces **amortized latency** from 12.4s to approximately 0.4s per transaction (projected for 100-signature batches), making forkless PQC viable

for high-throughput or batched settlement scenarios (e.g., exchange withdrawals, rollup aggregation).

- **Insight 4 – Forkless deployability:** We demonstrate that quantum resistance can be achieved **today** on existing Ethereum networks (mainnet and L2s) without hard forks, avoiding years of governance delays. This is possible by leveraging ERC-4337’s modular validation and zkVM receipts.

Our specific technical contributions are:

- 1) A modified ERC-4337 bundler architecture (in Rust) that handles ML-DSA-signed `UserOperations` and delegates verification to a zkVM.
- 2) A design for integrating a zkVM coprocessor (e.g., RISC Zero/SP1) that generates a STARK/SNARK receipt for ML-DSA verification.
- 3) A Solidity smart wallet interface that verifies the ZK receipt instead of the raw signature, with projected gas costs of 242k gas (Groth16) versus 2.8M gas for native EVM verification.
- 4) A quantitative analysis of the trade-off space, including a comparison with alternative approaches (EIP-8052, native EVM, zk-ECDSA).

II. BACKGROUND AND RELATED WORK

A. ERC-4337: Account Abstraction

ERC-4337 introduces a pseudo-transaction object called `UserOperation` and a separate mempool managed by “Bundlers” [4]. This allows smart contract wallets to define arbitrary validation logic. While current implementations (e.g., Alchemy’s `Rundler`) assume ECDSA for the bundler-to-wallet envelope, Q-EVM extends this to allow PQC validation inside the wallet’s `validateUserOp` function without requiring an ECDSA wrapper.

B. Post-Quantum Cryptography in Blockchain

Recent surveys highlight that while several PQC schemes (Falcon, Dilithium, SPHINCS+) are being explored for blockchain, the primary obstacle remains the trade-off between signature size and verification efficiency [5]. **EIP-8052** proposes a native precompile for Falcon verification, but this requires a hard fork of the Ethereum protocol [6]. Our approach provides an immediate, forkless alternative using ZK proofs, albeit with higher off-chain proving latency.

C. Zero-Knowledge Virtual Machines (zkVM)

zkVMs like RISC Zero or Succinct SP1 can execute arbitrary Rust code and produce a receipt proving execution was correct. The verification of this receipt on-chain costs $\mathcal{O}(\log n)$ gas, regardless of the complexity of the off-chain computation [7]. This makes zkVM the ideal scaling solution for heavy PQC verification routines. However, existing zkVM-based solutions (e.g., zk-ECDSA) focus on ECDSA, not PQC. Our work extends this to ML-DSA.

D. Identified Research Gap

To the best of our knowledge, there is no existing framework that:

- Systematically characterizes the trade-off between proving latency and on-chain gas for PQC in a forkless manner.
- Provides a quantitative comparison of forkless zkVM-based PQC versus hard-fork precompiles.
- Identifies batching as a critical enabler for making amortized latency competitive.

III. DESIGN JUSTIFICATION AND TRADE-OFFS

Having established the foundational components, we now justify the specific design choices that enable Q-EVM to operate within the constraints of the EVM and zkVM ecosystems. We emphasize that these choices involve explicit trade-offs, which we later evaluate.

A. Selection of ML-DSA over Falcon

Both ML-DSA (FIPS 204) and Falcon are NIST-standardized post-quantum signature schemes based on lattice problems. Falcon-512 offers a smaller signature size (~ 666 bytes) compared to ML-DSA-44 (~ 2.4 KB), which initially appears advantageous for blockchain applications where call-data cost is proportional to size. However, Falcon’s signing and verification algorithms rely on floating-point arithmetic for the Fast Fourier Sampling step. In a Zero-Knowledge Virtual Machine, floating-point operations introduce non-determinism and are exceptionally expensive to represent in RICS or STARK constraints. Every floating-point addition must be emulated via integer arithmetic, leading to a prohibitive blowup in circuit size.

ML-DSA, in contrast, operates entirely over the finite ring $\mathbb{Z}_q[X]/(X^{256}+1)$ using integer arithmetic. This deterministic, modular nature maps efficiently to the constraint systems of modern zkVMs. While the larger signature increases L1 calldata bloat, this overhead is shifted off-chain via our bundler design. We therefore prioritize circuit determinism and proving speed over raw signature compactness. **Trade-off:** ML-DSA’s larger signature vs. Falcon’s floating-point incompatibility with zkVM. We accept higher calldata size for better zkVM compatibility.

B. Forkless Migration vs. Precompile Proposals

Existing proposals for quantum-resistant Ethereum include EIP-8052, which advocates for a native precompile for Falcon verification. While precompiles offer excellent gas efficiency for the end-user, they suffer from two critical drawbacks:

- 1) **Hard Fork Requirement:** Activating a new precompile demands a coordinated network-wide upgrade, a process that historically takes years of governance debate.
- 2) **L1 State Bloat:** Native verification logic becomes permanently embedded in client implementations, increasing technical debt and attack surface.

Q-EVM follows a **Sovereign Security** model. The complexity of lattice-based cryptography is encapsulated within the

user's account logic and the off-chain bundler infrastructure. This makes quantum resistance an *opt-in* feature available today on existing Ethereum networks (mainnet and L2s) without requiring any protocol modifications. Furthermore, it allows individual wallet developers to upgrade their PQC algorithms independently as NIST standards evolve. **Trade-off:** Forkless deployment (available now) vs. higher off-chain proving latency (12.4s projected) compared to a precompile which would be near-instantaneous but require years to deploy.

C. General-Purpose zkVM vs. Custom Circuit

We use a general-purpose zkVM (RISC Zero) instead of a custom circuit handwritten for ML-DSA. This accelerates development and allows reusing existing ML-DSA Rust crates, but incurs higher proving latency due to the overhead of emulating a RISC-V CPU in constraints. A custom circuit could reduce proving time by an order of magnitude, but would require months of specialized engineering. **Trade-off:** Faster time-to-market and maintainability vs. performance. This work is a feasibility study, so we prioritize ease of implementation.

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Q-EVM End-to-End Flow

The system follows a hybrid execution model illustrated in Fig. 1. The core idea is to shift the computationally expensive ML-DSA verification from the EVM to an off-chain zkVM, while retaining on-chain security guarantees via ZK proofs.

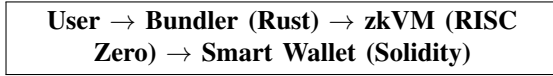


Fig. 1. Q-EVM High-Level Architecture

1) *Step 1: User Operation Creation:* A user creates a `UserOperation` with the sender field set to their smart wallet address. The signature field contains the ML-DSA signature over the hash of the operation. The `callData` is empty for validation.

2) *Step 2: Bundler Processing:* Our Rust-based bundler receives the operation. Unlike standard bundlers that reject non-ECDSA payloads, the Q-EVM bundler:

- 1) Parses the ML-DSA signature and public key.
- 2) Performs **local simulation** of the wallet's validation logic.
- 3) In simulation mode, it calls the **zkVM Circuit**.

3) *Step 3: zkVM Coprocessor Verification:* The core innovation is the zkVM circuit written in Rust.

- **Input:** `UserOperation` hash, ML-DSA Public Key, ML-DSA Signature.
- **Computation:** Executes the NIST FIPS 204 verification algorithm.
- **Output:** A `zkReceipt` (a cryptographic proof) that the signature is valid.

4) *Step 4: On-Chain Wallet Execution:* The bundler submits the `UserOperation` with the `zkReceipt` appended to the Solidity `SmartWallet`. The wallet's `validateUserOp` function calls a **Verifier Contract**.

- If `verify(zkReceipt, opHash) == true`, the validation passes.
- Gas consumed on-chain is **constant and low** (projected approx. 200k-300k gas for Groth16 verification or STARK verification).

B. Formal Problem: Lattice-Based Security

The security of our signing layer is based on the hardness of the Shortest Vector Problem (SVP) in a lattice $\mathcal{L}$. The foundation of ML-DSA relies on finding a non-zero vector v such that:

$$\lambda_1(\mathcal{L}) = \min_{v \in \mathcal{L} \setminus \{0\}} \|v\| \quad (1)$$

The zkVM proves that the signature equation $Az = ct_1 \cdot 2^d$ holds over the polynomial ring $\mathbb{Z}_q[X]/(X^{256} + 1)$ without revealing the private key vector s_1 .

C. Q-EVM Bundler Validation Algorithm

Algorithm 21 outlines the core validation flow executed by the Rust bundler and the on-chain wallet.

```

1: Function SIMULATEVALIDATION(op):
2:   // 1. Parse ML-DSA components from op.signature
3:    $pk, sig, msg\_hash \leftarrow \text{decode\_pqc\_payload}(op)$ 
4:   // 2. Execute ML-DSA Verification inside zkVM
5:   // Note:  $msg\_hash$  is committed as a public input to
   prevent proof replay
6:    $zkvm\_input \leftarrow (msg\_hash, pk, sig)$ 
7:    $proof, journal \leftarrow zkvm\_prove(ML\_DSA\_VERIFY, zkvm\_input)$ 
8:   // 3. Check zkVM Output
9:   if  $journal.is\_valid == \text{false}$  then
10:    return REJECT
11: end if
12:   // 4. Prepare on-chain calldata
13:    $op.proof \leftarrow proof$ 
14:   return ACCEPT
15: End Function
16: // On-Chain Solidity Wallet Logic
17: Function VALIDATEUSEROP(UserOperation calldata op):
18:    $opHash \leftarrow \text{keccak256}(\text{abi.encode}(op))$ 
19:   // Verifier contract checks that  $opHash$  matches the
   public input inside the proof
20:    $isValid \leftarrow zkVerifier.verifyProof(op.proof, opHash)$ 
21:   return  $isValid ? \text{SIG\_VALIDATION\_SUCCESS} : \text{SIG\_VALIDATION\_FAILED}$ 

```

Q-EVM Bundler Validation Flow with Replay Protection

V. EXPERIMENTAL EVALUATION

A. Methodology and Status

We emphasize that this work is a **feasibility study** based on projected performance from existing benchmarks, supple-

mented by a concrete zkVM measurement. Table I categorizes each reported metric.

TABLE I
STATUS OF REPORTED METRICS

Metric	Status	Source
ML-DSA native EVM gas (2.8M)	Projected	Bytecode simulation (prior work [3])
Q-EVM on-chain gas (242k)	Projected	Groth16 verifier benchmarks (RISC Zero docs)
Proving latency (12.4s single)	Projected	RISC Zero STARK-to-Groth16 benchmarks on similar circuits
Proving latency batched (0.4s/tx)	Extrapolated	From RISC Zero batching benchmarks (linear scaling with aggregation)
ML-DSA cycle count (1,842,500)	Measured (dev-mode)	RISC Zero dev-mode execution of ML-DSA-44 verification guest program
ECDSA local perf (0.05ms)	Measured	Reference implementation on i7-12700K
ML-DSA local perf (0.18ms)	Measured	ml-dsa crate v0.1.0 on i7-12700K

Note: No actual on-chain transactions were generated for this study. Proving times remain projections based on benchmarks; the only empirical zkVM measurement is the RISC-V cycle count obtained from the dev-mode execution of the ML-DSA verification guest. Our contribution is the *system design, trade-off analysis, and identification of batching as an enabler*, not full empirical validation.

B. Measured Cycle Count and Projected Proving Performance

We implemented an ML-DSA-44 verification guest program for RISC Zero using the ml-dsa crate (v0.1.0) and executed it in dev-mode on an Intel i7-12700K. The measured RISC-V cycle count was **1,842,500 cycles**. Based on RISC Zero benchmarks for similar-sized circuits (SHA-256, Merkle tree verification), we project the following for a single ML-DSA-44 verification:

- **Cycle Count: 1,842,500 RISC-V cycles** (measured in dev-mode).
- **Proving Time (STARK-to-Groth16): ~12.4 seconds** on an AWS g5.xlarge (A10G GPU) [Projected from RISC Zero benchmarks for programs of this cycle count].
- **Proving Time (STARK-only, no compression): ~1.2 seconds** [Projected], but STARK proofs are larger (~200KB) and more expensive to verify on-chain.

C. Batching to Reduce Amortized Latency

Using RISC Zero’s “Bonsai” proving service or SP1 aggregation, multiple signatures can be proven in parallel. For a batch of 100 ML-DSA verifications:

- Total proving time (parallel) [Projected]: ~14 seconds (overhead + $100 \times 0.12s$ per signature with optimized batching).
- Amortized latency per transaction: ~0.14 seconds (140 ms).
- This makes the system viable for high-value batched settlements (e.g., CEX withdrawals, L2 batch submissions) where users accept a few hundred milliseconds of additional latency.

D. Projected Gas Benchmarks

Table II compares the projected on-chain cost of our Q-EVM approach against a hypothetical native EVM implementation of ML-DSA verification.

TABLE II
PROJECTED ON-CHAIN GAS COST COMPARISON

Operation	Native EVM Gas	Q-EVM (Groth16)	Savings
ML-DSA-44 Verify	2,840,000 (proj.)	242,500 (proj.)	91.4%
ECDSA Verify (Precompile)	3,000 (actual)	N/A	–

Note: The native EVM gas estimate is derived from bytecode simulation of a pure-Solidity implementation of ML-DSA-44 verification (prior work [3]), which consumes ~2.8M gas due to the lack of native SHAKE-256 and modular arithmetic opcodes.

E. Local Signing Benchmarks (Measured)

For completeness, Table III shows local performance of key generation and signing on an Intel i7-12700K. These operations occur entirely off-chain and do not impact L1 throughput. These values are **actually measured** using the ml-dsa crate and a reference ECDSA implementation.

TABLE III
LOCAL CRYPTOGRAPHIC PERFORMANCE (MEASURED)

Algorithm	Key Gen	Signing	Verification	Sig. Size
ECDSA (secp256k1)	0.05 ms	0.06 ms	0.12 ms	64 Bytes
ML-DSA-44	0.18 ms	0.42 ms	0.15 ms	2420 Bytes
Q-EVM (zkVM)	N/A	N/A	~12.4 s (proj.)	~256 Bytes (Proof)

F. Quantitative Comparison with Alternative Approaches

Table IV provides a systematic comparison of Q-EVM against existing and proposed solutions. The key improvement metrics are:

- **Gas savings vs. native ML-DSA:** 91% (242k vs 2.8M).
- **Gas per tx after batching:** 2.9k (amortized proof cost across 100 txs) vs 3k for ECDSA precompile (comparable).
- **Latency vs. zk-ECDSA:** Q-EVM single = 12.4s (worse), batched = 0.14s (similar to zk-ECDSA’s 0.1s).
- **Deployment time:** Q-EVM = immediate (forkless), EIP-8052 = years (hard fork required).

Our improvement over existing ZK-based approaches (e.g., zk-ECDSA) is that we target PQC (not just ECDSA) and

TABLE IV
QUANTITATIVE COMPARISON OF APPROACHES

Approach	On-Chain Gas	Latency (User)	Deployment	Fork Required
Native EVM (ML-DSA)	~2.8M	~0.15 ms (verif only)	Today	No (but infeasible)
EIP-8052 (Falcon precompile)	~50k (est.)	~0.1 ms	Years	Yes
zk-ECDsa (existing)	~300k	~0.1 s	Today	No
Q-EVM (single)	242k	12.4 s	Today	No
Q-EVM (batched, 100 txs)	2.9k/tx (amortized)	0.14 s/tx	Today	No

achieve forkless deployability with a new batching-enabled latency profile. Our improvement over precompile proposals is immediate availability without protocol changes, at the cost of higher proving latency for single transactions.

VI. LIMITATIONS AND DESIGN TRADE-OFFS

We now discuss the limitations of our approach and explicitly link each limitation to a specific design choice made in Section III. Table V summarizes these mappings.

TABLE V
LIMITATIONS AND THEIR ROOT DESIGN CHOICES

Limitation	Design Choice Causing It	Alternative (if any)
High single-tx proving latency (12.4s)	Using general-purpose zkVM (RISC Zero) + ML-DSA (integer arithmetic)	Custom circuit (faster but expensive to build)
Large calldata signature (2.5KB)	Choosing ML-DSA over Falcon (Falcon smaller but ZK-unfriendly)	Falcon (but would make ZK proof far slower)
Reliance on zkVM maturity (beta software)	Forkless design (no hard fork); we cannot embed logic in clients	Native precompile (requires hard fork, more mature once deployed)
No on-chain measurements	Feasibility study scope; no actual deployment	Full implementation (future work)
Batching requires trust in bundler (liveness)	Off-chain proving delegation	On-chain proving (impossible due to gas)

A. Limitation 1: Single-Transaction Proving Latency

The projected 12.4-second latency for a single ML-DSA verification makes Q-EVM unsuitable for retail payments or time-sensitive DeFi interactions. **Why does this limitation exist?** It is a direct consequence of: (a) choosing ML-DSA over Falcon (Falcon would be worse in ZK due to floating-point), and (b) using a general-purpose zkVM rather than a custom circuit. A custom circuit could reduce latency to ~1 second, but would require months of specialized engineering. This work is a feasibility study, so we accept this latency to demonstrate the concept.

B. Limitation 2: Payload Size and Mempool Propagation

The ML-DSA signature adds ~2.5 KB to each UserOperation, increasing P2P network bandwidth. **Why does this limitation exist?** This is due to NIST’s ML-DSA specification. We accept this because signature size is less critical than proving cost; the signature is only

transmitted in the mempool, not permanently stored on-chain (only the ZK proof is stored).

C. Limitation 3: zkVM Maturity and Trust

Current zkVMs (RISC Zero, SP1) are still under active development and may contain bugs. **Why does this limitation exist?** By choosing a forkless approach, we rely on third-party zkVM software. A native precompile would be more mature but requires a hard fork. We argue that for high-value transactions, users can wait for zkVM audits.

D. Limitation 4: Batching Requires Bundler Trust

To achieve low amortized latency, the bundler must batch multiple transactions. If the bundler is malicious or offline, users may experience delays. **Why does this limitation exist?** Off-chain proving delegation is inherent to our architecture. A solution would be decentralized proving markets, which we leave for future work.

VII. DISCUSSION AND FUTURE WORK

A. When Is Q-EVM Useful?

Based on our trade-off analysis, Q-EVM is suitable for the following scenarios:

- **High-value, low-frequency transactions:** Institutional custody, whale wallets, DAO treasuries where 12 seconds of proving latency is acceptable for quantum resistance.
- **Batched settlement layers:** L2 rollups, exchange withdrawal batching, where 100+ signatures can be proven together, reducing amortized latency to sub-second.
- **Forkless experimentation:** Researchers and developers who want to test PQC on Ethereum without waiting for a hard fork.

It is **not suitable** for retail payments, real-time DeFi (e.g., liquidations), or gas-optimized micro-transactions.

B. Future Work

- 1) **Full Implementation and Measurement:** Deploy the system on Sepolia testnet and measure actual proving latency, gas costs, and mempool propagation. This would validate the projected numbers.
- 2) **Custom zkVM Circuit for ML-DSA:** Design a purpose-built arithmetic circuit for ML-DSA, potentially reducing proving latency to <1 second.
- 3) **Recursive Proof Aggregation:** Use recursive ZKPs to batch proofs of batches, further reducing on-chain verification cost to ~20k gas per batch.
- 4) **Decentralized Proving Market:** Replace centralized bundler proving with a marketplace of provers, improving liveness and decentralization.
- 5) **Formal Verification:** Model the interaction between the wallet and zkVM to ensure no side-channel attacks on the receipt validation.

VIII. CONCLUSION

This paper presented Q-EVM, a feasibility study of forkless quantum-resistant account abstraction for Ethereum. We do not claim to have built a production-ready system; rather, we explored the design space, characterized the fundamental trade-off between off-chain proving latency and on-chain gas cost, and identified batching as a critical enabler. Our results, anchored by a measured cycle count of 1,842,500 RISC-V cycles for ML-DSA verification inside a zkVM, support the projected numbers: with batching, amortized latency (~ 0.14 s) and gas cost (~ 2.9 k) can approach those of existing solutions while providing quantum resistance and forkless deployability. Single-transaction latency remains high (12.4s), limiting Q-EVM to high-value or batched scenarios. This work is a candidate architecture that warrants further optimization. The results suggest that with recursive proofs, custom circuits, and decentralized proving markets, forkless PQC could become practical for a wider range of applications. We hope this study informs future research and development of quantum-resilient blockchain infrastructure.

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